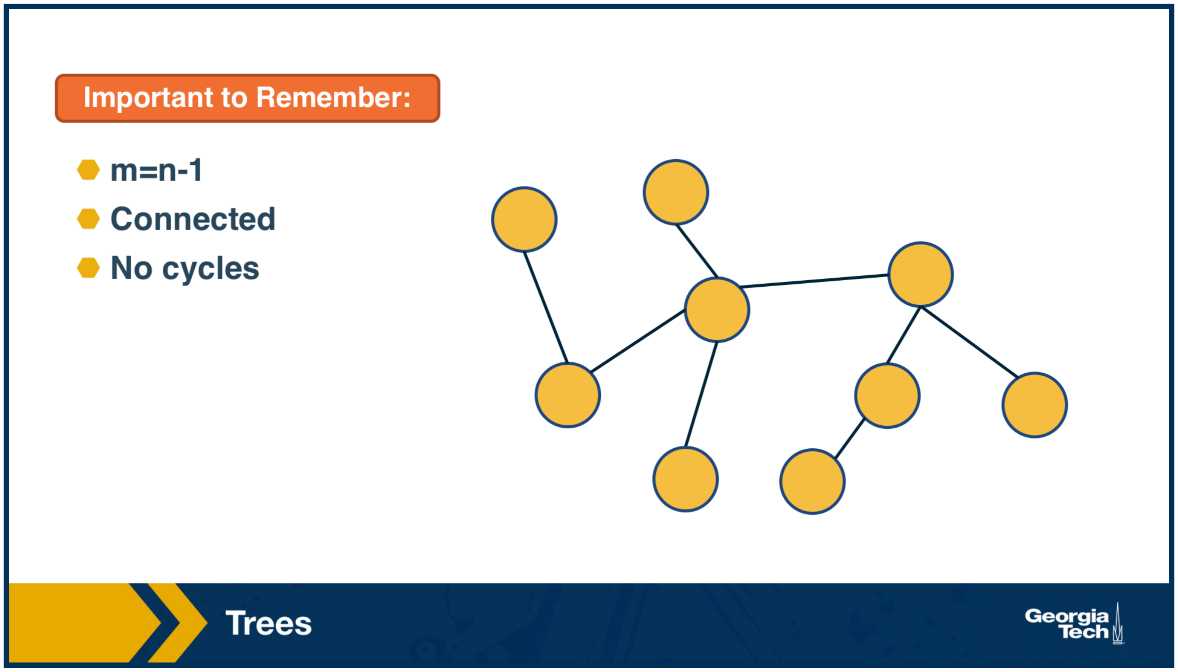
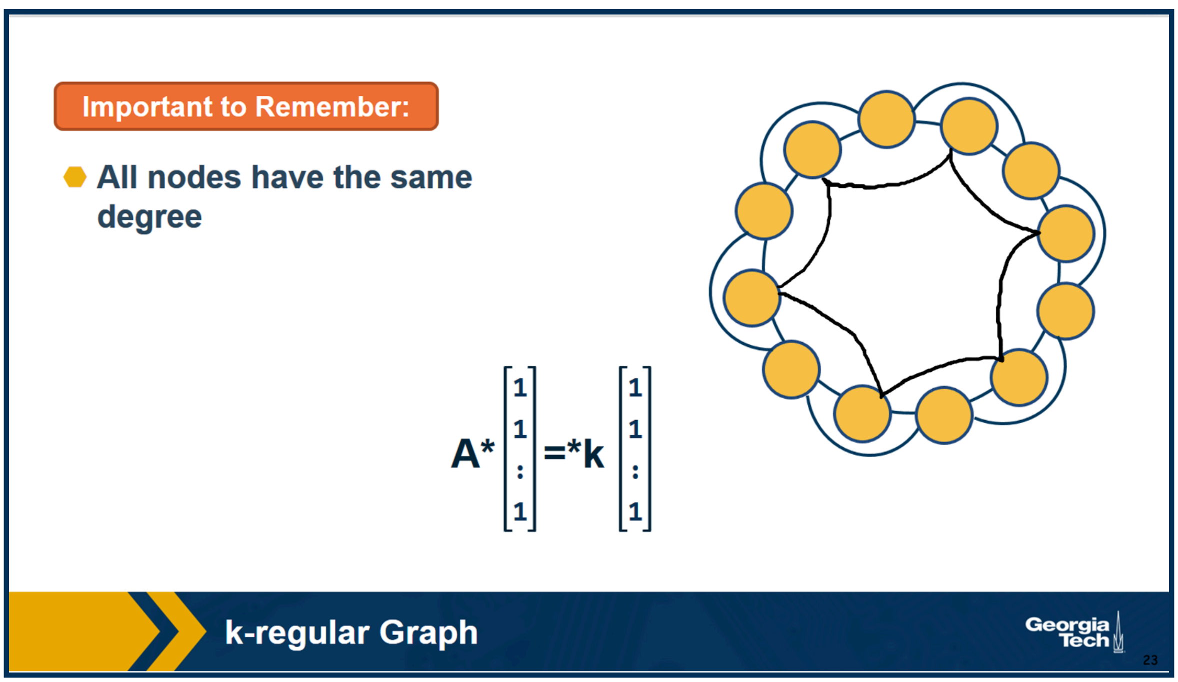
L2: Trees and Other Regular Networks

In graph theory, the focus is often on some special classes of networks, such as trees, lattices, regular networks, planar graphs, etc.

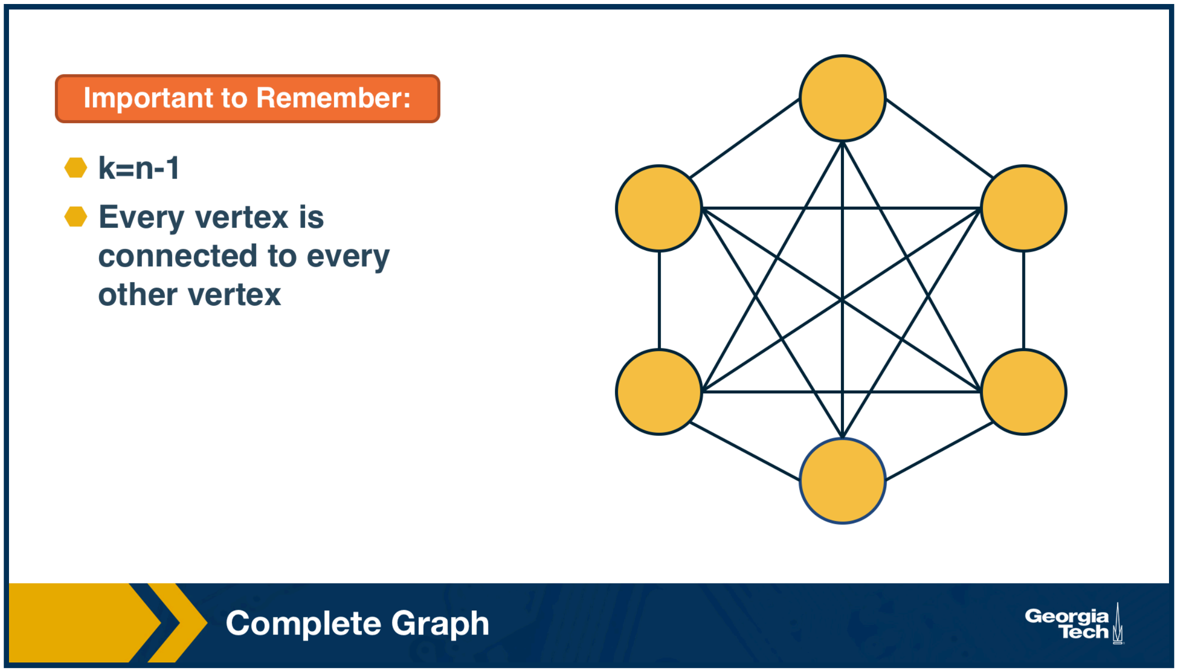
In this course, we will focus instead on complex graphs that do not fit in any of these special classes. However, we will sometimes contrast and compare the properties of complex networks with some regular graphs.



For instance, trees are connected graphs that do not have cycles – and you can easily show that the number of edges in a tree of n nodes is always m=n-1.



A k-regular graph is a network in which every vertex has the same degree k. The visualization shows an example of a k-regular network for k=4.



A complete graph *(or "clique")* is a special case of a regular network in which every vertex is connected to every other vertex (k=n-1). The example shows a clique with 6 nodes.

Food for Thought

Suppose that a graph is k-regular. How would you show that a vector of n ones *(1, 1, … 1)* is an eigenvector of the adjacency matrix -- and the corresponding eigenvalue is equal to k?